

EDC Gravity Sector (BLOCK-003):

A 5D-First Derivation Program

From Bulk+Brane+GHY+Israel to an Effective 4D Newton Coupling

EDC Research Program

Status: **[OPEN]** — *Derivation roadmap, not results*

2026-02-02

Program note / roadmap; not a results paper

Abstract

BLOCK-003 designates the open problem of deriving the 4D Newton coupling G_N within the Elastic Diffusive Cosmology (EDC) framework without circular input. This note provides a structured derivation program: starting from the 5D bulk Einstein–Hilbert action with Gibbons–Hawking–York boundary terms and brane tension, through Israel junction conditions, to extraction of an effective 4D gravitational coupling. A central requirement is *anti-circularity*: inputs used here (bulk parameters κ_5 , Λ_5 , σ) must be independent of those already employed in EM/nuclear sector closures. An explicit Input Ledger separates allowed from prohibited inputs. No numerical closure is claimed; no fit to G_N^{obs} is performed. Closure would require deriving the correct scaling $G_N \sim \kappa_5^2/R_\xi$ (or equivalent) and recovering the 4D Newtonian limit at large distances. This note is a standalone companion to the EDC nuclear pinning monograph, which remains gravity-out-of-scope.

1 Epistemic Status & Scope

Epistemic Status

Status: **[OPEN]**/**[P]**. This paper is a *derivation roadmap*, not a completed result. The gravity sector (BLOCK-003) remains open. No numerical predictions for G_N or gravitational signatures are made here.

What this paper is. A structured research program for deriving 4D gravity from a 5D brane-world setup consistent with EDC conventions. The derivation chain is outlined; key steps are identified; acceptance criteria are defined.

What this paper is not.

- Not a claim that gravity is “solved” in EDC.
- Not a source of new physics claims or predictions.
- Not a replacement for the nuclear pinning monograph (which remains gravity-out-of-scope).

Epistemic tags used.

- **[BL]**: Standard brane-world formalism (textbook results).
- **[I]**: Identification or interpretation within EDC framework.
- **[P]**: EDC-specific postulate (not independently derived).
- **[OPEN]**: Open problem; derivation not complete.

Notation summary.

Symbol	Meaning
$X^A = (w, x^i, \xi)$	Bulk coordinates ($A = 0, \dots, 4$)
w	Bulk time
t	Brane operational time; $w = v_{\text{scan}} t$
Σ^4	4D brane hypersurface
σ	Brane tension
$\kappa_5^2 = 8\pi G_5$	5D gravitational coupling
Λ_5	Bulk cosmological constant
$h_{\mu\nu}, K_{\mu\nu}$	Induced metric, extrinsic curvature

2 Canonical 5D Geometric Setup (EDC-Consistent)

Bulk coordinates [BL]/[I]. EDC postulates a 5D Lorentzian bulk manifold $\mathcal{M}^5$ with coordinates

$$X^A = (w, x^1, x^2, x^3, \xi), \quad A = 0, \dots, 4, \quad (1)$$

where w is the bulk time coordinate and ξ is the extra spatial coordinate.

Brane and operational time [I]. A 4D brane Σ^4 is embedded in $\mathcal{M}^5$. Brane observers measure operational time t , related to bulk time via the scan map $w = v_{\text{scan}} t$. In many applications one sets $v_{\text{scan}} = 1$; the conceptual distinction is retained.

Induced metric [BL]. The brane inherits an induced metric from the bulk:

$$h_{\mu\nu}(x) = \partial_\mu X^A \partial_\nu X^B g_{AB}(X)|_{\Sigma}, \quad \mu, \nu = 0, \dots, 3. \quad (2)$$

Metric ansatz (optional) [P]. A specific warped metric (e.g., $ds_5^2 = e^{-2A(\xi)} \eta_{\mu\nu} dx^\mu dx^\nu + d\xi^2$) may be adopted for concrete calculations. Such an ansatz is *not* prescribed here as “the” EDC metric; any choice must be labeled [P] and justified separately.

3 Action Principle (5D-First)

The total action consists of three parts [BL] (see [1, 2] for standard GR formulation):

5D Action Components

1. Bulk Einstein–Hilbert + cosmological term:

$$S_{\text{bulk}} = \frac{1}{2\kappa_5^2} \int_{\mathcal{M}^5} d^5 X \sqrt{-g^{(5)}} \left(R^{(5)} - 2\Lambda_5 \right) \quad (3)$$

2. Gibbons–Hawking–York boundary term:

$$S_{\text{GHY}} = \frac{1}{\kappa_5^2} \int_{\Sigma^4} d^4 x \sqrt{-h} K \quad (4)$$

3. Brane action (tension + matter):

$$S_{\text{brane}} = \int_{\Sigma^4} d^4 x \sqrt{-h} (-\sigma + \mathcal{L}_{\text{matter}}) \quad (5)$$

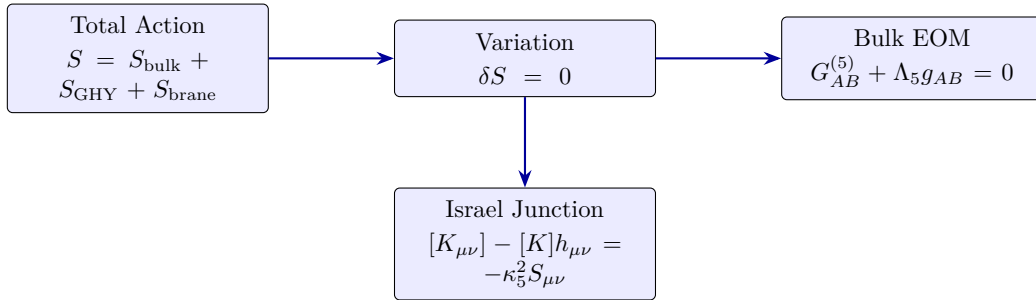
Notation.

- $\kappa_5^2 = 8\pi G_5$ is the 5D gravitational coupling.
- Λ_5 is the bulk cosmological constant.
- $K = h^{\mu\nu} K_{\mu\nu}$ is the trace of the extrinsic curvature.
- σ is the brane tension.

Why GHY is required [BL]. The GHY term [3, 4] is necessary for a well-posed variational principle when the manifold has boundaries. Without it, the variation of $R^{(5)}$ produces boundary terms that do not vanish, making the action principle ill-defined.

4 Junction Conditions & Field Equations

Derivation chain (outline) [BL].



Israel junction conditions [BL]. At the brane location, the discontinuity in extrinsic curvature is related to the brane stress-energy [5]:

$$[K_{\mu\nu}] - [K]h_{\mu\nu} = -\kappa_5^2 \left(T_{\mu\nu}^{\text{brane}} - \frac{1}{3} T h_{\mu\nu} \right) \quad (6)$$

For a tension-dominated brane ($T_{\mu\nu} = -\sigma h_{\mu\nu}$), this becomes:

$$[K_{\mu\nu}] = -\frac{\kappa_5^2}{3} \sigma h_{\mu\nu} + \kappa_5^2 \left(\tau_{\mu\nu} - \frac{1}{3} \tau h_{\mu\nu} \right) \quad (7)$$

where $\tau_{\mu\nu}$ is the matter stress-energy on the brane (excluding tension).

Status. Equations (6) are standard results [BL]. Their application to a specific EDC metric ansatz requires further work [OPEN].

5 Linearized Gravity & Effective 4D Newton Coupling (Program)

Epistemic Status

Status: [OPEN]. This section outlines the *required steps* to extract G_N . No calculation is performed here.

Goal. Obtain an effective 4D gravitational potential on Σ^4 with $1/r$ long-distance behavior and identify G_N in terms of 5D parameters.

Required steps.

- Step 1: Choose background.** Select a specific 5D metric ansatz (flat, warped, or EDC-specific) and solve for the background geometry.
- Step 2: Linearize.** Expand $g_{AB} = g_{AB}^{(0)} + h_{AB}$ and derive linearized field equations.
- Step 3: Mode decomposition.** Perform Kaluza–Klein decomposition; identify tensor modes (4D graviton), scalar modes (radion/brane-bending), and their mass spectra.
- Step 4: Normalization.** Normalize mode functions with respect to the appropriate inner product to obtain canonical kinetic terms.
- Step 5: Brane-to-brane propagator.** Compute the Green function for metric perturbations sourced and observed on the brane.
- Step 6: Extract G_N .** From the long-distance ($r \gg$ compactification scale) limit of the brane-to-brane potential, identify the coefficient as G_N .

Expected form (schematic) [OPEN]. In typical brane-world scenarios [6, 7], the effective Newton constant takes the form:

$$G_N \sim \frac{\kappa_5^2}{R_\xi} \quad \text{or} \quad G_N \sim \kappa_5^2 \cdot (\text{warp factor integral}) \quad (8)$$

The precise form depends on the metric ansatz and must be derived, not assumed.

6 Input Ledger (Anti-Circularity)

Warning: Anti-Circularity

To avoid circular reasoning, we must distinguish:

- **Allowed inputs:** Parameters that can be used to derive G_N without circularity.
- **Prohibited inputs:** Parameters already used to close the EM/nuclear sectors (e.g., α, r_e).

Using prohibited inputs to derive G_N would make gravity a consistency check, not a prediction.

Quantity	Status	Source / Notes
κ_5 or M_5	[OPEN]	5D Planck mass; must be independently fixed or predicted.
Λ_5	[P]	Bulk cosmological constant; may be set by consistency (e.g., flat brane requirement).
R_ξ	[P]/[OPEN]	Compactification scale; geometry parameter. If derived from σ , see below.
σ (intrinsic)	[P]	Brane tension as bulk property. Allowed: if derived from $\sigma = 2\pi R_\xi^2 \rho_P$ (pressure integration).

Quantity	Status	Source / Notes
σ_{eff}	[I]	Effective tension at EM scale. Prohibited: if extracted from α , r_e (circularity risk).
α	[BL]	Fine structure constant. Prohibited for gravity derivation (used in EM closure).
r_e	[BL]	Classical electron radius. Prohibited for gravity derivation (used in EM closure).
m_e	[BL]	Electron mass. Prohibited for gravity derivation (used in EM closure).
Warp factor $A(\xi)$	[P]/[OPEN]	If used, must be derived or postulated independently.

Guiding principle. For G_N to qualify as a *prediction* of EDC (not merely a consistency check), it must be derivable from:

- Geometric parameters (R_ξ , warp profile) determined by 5D dynamics, OR
- The intrinsic tension σ derived from bulk pressure integration (non-circular path).

7 Pipeline Schematic

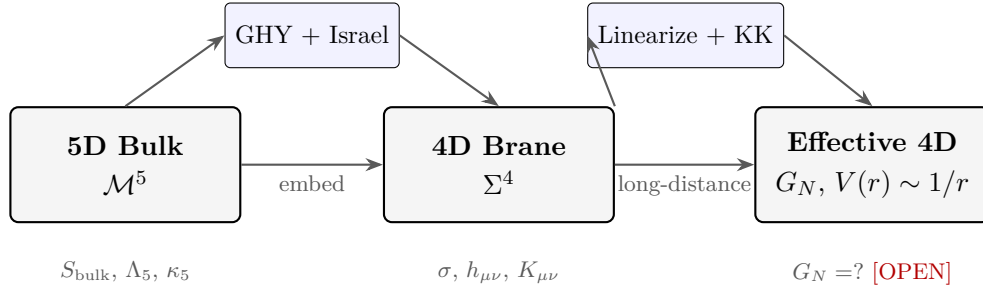


Figure 1: Schematic derivation pipeline for BLOCK-003 (program overview). Starting from the 5D bulk action, apply GHY and Israel junction conditions at the brane, linearize, and extract the effective 4D Newton coupling G_N . Status: [OPEN].

8 Connection to Nuclear Pinning Monograph

This paper is a *standalone companion* to the EDC nuclear pinning monograph (“Topological Pinning Model for Nuclear Structure and Superheavy α -Decay Predictions”).

Separation of concerns.

- The nuclear monograph addresses nuclear binding, α -decay, and topological pinning. Gravity is explicitly out-of-scope there (BLOCK-003).
- This paper addresses the gravity derivation program. Nuclear physics is not treated here.

No modifications to the monograph. This paper does not add, remove, or modify any content in the nuclear monograph. The monograph may contain a single-sentence pointer: “See separate gravity program note for BLOCK-003.” No other cross-references are required.

9 Roadmap & Acceptance Criteria

The following acceptance criteria define what is needed to close BLOCK-003:

ID	Criterion	Status
AC-G1	Well-posed 5D action with bulk, GHY, and brane terms explicitly written.	[BL] Done
AC-G2	Israel junction conditions derived or correctly quoted with proper index structure.	[BL] Done
AC-G3	Specific metric ansatz chosen and background solution obtained.	[OPEN]
AC-G4	Linearized perturbation equations derived around background.	[OPEN]
AC-G5	KK modes identified, classified (tensor/scalar), and normalized.	[OPEN]
AC-G6	Brane-to-brane propagator computed; long-distance limit shows $1/r$ behavior.	[OPEN]
AC-G7	G_N extracted and expressed in terms of 5D parameters without circularity.	[OPEN]
AC-G8	Consistency check: G_N value compatible with measured Newton constant (order of magnitude or better).	[OPEN]
AC-G9	Parameter counting: clearly state what is input vs. what is predicted.	[OPEN]
AC-G10	Reproducibility: derivation steps documented such that an independent researcher can verify.	[OPEN]

Current status. Only AC-G1 and AC-G2 are addressed in this document (as standard results). All other criteria remain [OPEN].

10 Conclusion

This document outlines a structured 5D-first research program for BLOCK-003 (gravity sector) within EDC. The main elements are:

1. A derivation roadmap from 5D action to effective 4D G_N .
2. An explicit *input ledger* to prevent circular reasoning.
3. Defined acceptance criteria (AC-G1 through AC-G10) for closure.

No physics results or numerical predictions are claimed. This note is intended as a starting point for future work on the EDC gravity sector.

References

- [1] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation* (W. H. Freeman, San Francisco, 1973).
- [2] R. M. Wald, *General Relativity* (University of Chicago Press, 1984).
- [3] J. W. York, “Role of conformal three-geometry in the dynamics of gravitation,” *Phys. Rev. Lett.* **28**, 1082 (1972).
- [4] G. W. Gibbons and S. W. Hawking, “Action integrals and partition functions in quantum gravity,” *Phys. Rev. D* **15**, 2752 (1977).
- [5] W. Israel, “Singular hypersurfaces and thin shells in general relativity,” *Nuovo Cimento B* **44**, 1 (1966).
- [6] L. Randall and R. Sundrum, “Large mass hierarchy from a small extra dimension,” *Phys. Rev. Lett.* **83**, 3370 (1999).
- [7] R. Maartens and K. Koyama, “Brane-world gravity,” *Living Rev. Relativ.* **13**, 5 (2010).